08

Lecture 08

Random Access Channel (RACH)

کانال دسترسی تصادفی (RACH)

2G — 3G — **4G** — 5G — 6G

COURSE PART
Part II – 4G access and core (Lectures 5–13)

ESTIMATED STUDY TIME
6–8 hours (3 h reading, 3–5 h exercises)

DIFFICULTY
Intermediate

PREREQUISITES
Lectures 6–7; slotted ALOHA; complex exponentials and correlation

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Original lecture notes for the course *Advanced Communication Networks*. The lecture sequence follows the course syllabus (26 lectures, 2G to 6G). Equations use identical notation in the English and Persian editions. Companion edition: Persian (`lecture-08-fa.pdf`).

1 Learning objectives

After studying this lecture you should be able to:

- **LO1** **List** the situations in which an LTE UE must perform random access and **explain** why uplink synchronisation is the core problem.
- **LO2** **Describe** the PRACH: preamble formats, cyclic prefix, guard time and the maximum cell range they imply.
- **LO3** **Explain** Zadoff–Chu sequences, their correlation properties and how 64 preambles are generated per cell with cyclic shifts.
- **LO4** **Trace** contention-based (4-step) and contention-free random access, and the NR 2-step variant.
- **LO5** **Compute** timing advance values and the cell radius supported by a cyclic-shift configuration.
- **LO6** **Model** preamble collisions and RACH overload, and **evaluate** countermeasures such as access class barring and backoff.

2 Prerequisite knowledge

You need the LTE frame structure and uplink scheduling of Lecture 7 (in particular that PUSCH transmissions of different UEs must arrive time-aligned at the eNB), the timing advance concept of GSM (Lecture 1), the slotted ALOHA model of random access from introductory networking, and the definition of the periodic cross-correlation of two complex sequences.

Key terms. The vocabulary of this lecture with its Persian equivalents:

Term	Persian	Meaning in this lecture
Random access (RA)	دسترسی تصادفی	Uncoordinated first uplink access of a UE.
PRACH	کانال فیزیکی دسترسی تصادفی	Time–frequency resources reserved for preambles.
Preamble	پیش‌نوا	Known sequence sent by the UE in Msg1.
Zadoff–Chu (ZC) sequence	دنبالهٔ زادوف--چو	Constant-amplitude sequence with ideal cyclic autocorrelation.
Cyclic shift (N_{CS})	جابه‌جایی چرخشی	Shift that creates orthogonal preambles from one root.
Timing advance (TA)	پیش‌فرستی زمانی	Uplink advance compensating the round-trip delay.
RAR	پاسخ دسترسی تصادفی	Msg2: TA, uplink grant and temporary identity.
Contention resolution	حل رقابت	Msg4: tells which of the colliding UEs won.
Backoff	عقب‌نشینی	Random wait before a new attempt.
Access class barring (ACB)	مسدودسازی ردهٔ دسترسی	Probabilistic admission control before RA.

3 Introduction and motivation

Scheduled access needs two things the network does not have for a newly arriving UE: an *identity* to address it on the PDCCH and *uplink time alignment*. A UE that has just woken up knows the downlink timing (from the synchronisation signals) but not its distance to the eNB. If it simply transmitted on PUSCH, its symbols would arrive late by the round-trip delay and destroy the orthogonality of SC-FDMA with other users.

Random access is the procedure that solves both problems at once. The UE sends a short, robust signal (a *preamble*) in resources reserved for this purpose; the eNB measures its arrival time, sends back a timing correction and a first grant, and the two sides then resolve the possibility that several UEs chose the same preamble. The procedure is used surprisingly often – at every call set-up from idle, at every handover, after radio link failure and whenever an out-of-sync UE has data – so its capacity, latency and robustness matter. With massive IoT and 5G beam-based access, random access has become one of the most actively engineered parts of the radio interface.

4 Fundamental concepts

4.1 When random access is triggered

In LTE the MAC performs random access [1] for:

- initial access from RRC_IDLE (RRC connection establishment);
- RRC connection re-establishment after radio link failure;
- handover, to synchronise with the target cell;
- downlink data arrival for a connected UE that has lost uplink synchronisation;
- uplink data arrival when the UE is out of sync or has no scheduling-request resources;
- positioning, where the timing advance is needed.

4.2 Contention-based versus contention-free

In **contention-based** random access (CBRA) the UE picks a preamble at random from a common set; two UEs may pick the same one, so a contention-resolution step is needed. In **contention-free** random access (CFRA) the eNB assigns a dedicated preamble in advance (for handover or downlink data arrival); no collision can occur and the procedure ends after the RAR.

4.3 Zadoff–Chu sequences

Definition 1 • Zadoff–Chu sequence

For an odd prime length N_{ZC} and a root index $u \in \{1, \dots, N_{\mathrm{ZC}} - 1\}$,

$$x_u(n) = \exp\!\left(-\mathrm{j}\,\frac{\pi\, u\, n\,(n+1)}{N_{\mathrm{ZC}}}\right), \qquad n = 0, 1, \dots, N_{\mathrm{ZC}} - 1. \tag{1}$$

Its amplitude is constant, its periodic autocorrelation is zero for every non-zero shift, and the cross-correlation of two different roots has constant magnitude $1/\sqrt{N_{\mathrm{ZC}}}$ (normalised).

These properties make ZC sequences ideal preambles: constant amplitude means a low peak-to-average power ratio; zero autocorrelation means that *cyclic shifts* of one root are orthogonal and the delay of a received preamble shows up as a clean correlation peak. LTE uses $N_{\mathrm{ZC}} = 839$ for preamble formats 0–3 [2]; the script `scripts/validate_examples.py` verifies the properties numerically for $u = 25$ and $u = 29$.

5 Detailed technical explanation

5.1 The PRACH

The PRACH occupies six resource blocks (1.08 MHz) in the uplink, in subframes given by the *PRACH configuration index* broadcast in SIB2. Inside this band the preamble uses a subcarrier spacing of 1.25 kHz, so its 839 subcarriers fit in about 1.05 MHz. A preamble consists of a cyclic prefix, the ZC sequence (800 μs, or 1600 μs repeated) and an implicit guard time: Formats with longer CP and guard serve large cells; formats with a repeated sequence collect more energy for coverage-limited cells. Format 4 (TDD only) uses a short sequence of length 139 in the special subframe.

Table 1. LTE FDD preamble formats [2]. The guard time bounds the round-trip delay, hence the cell range $d_{\max} = c\,T_{\mathrm{GT}}/2$.

Format	Subframes	CP [μs]	Sequence [μs]	Guard [μs]	Range [km]
0	1	103.1	800	96.9	14.5
1	2	684.4	800	515.6	77.3
2	2	203.1	1600	196.9	29.5
3	3	684.4	1600	715.6	107.3

5.2 64 preambles per cell

Each cell offers 64 preambles. They are generated from one or more ZC roots by cyclic shifts of N_{CS} samples: one root yields $\lfloor N_{\mathrm{ZC}}/N_{\mathrm{CS}} \rfloor$ preambles, and further roots (consecutive in a broadcast logical order) are used until 64 are obtained. The shift must exceed the maximum round-trip delay plus delay spread; otherwise a distant UE's preamble would be detected as a neighbouring shift (Section 6.2). The 64 preambles are divided into contention-based groups (A and B, the latter signalling a larger Msg3) and a set reserved for contention-free use.

5.3 Contention-based random access (4 steps)

1. **Msg1 – preamble.** The UE selects a preamble at random from its group and transmits it in the next PRACH occasion, with an open-loop power estimate from the downlink path loss.
2. **Msg2 – random access response (RAR).** Within a response window the eNB sends, on PDSCH addressed by an RA-RNTI derived from the PRACH occasion, the detected preamble index (RAPID), a timing advance command (11 bits), an uplink grant for Msg3 and a temporary C-RNTI. If no RAR arrives, the UE *ramps* its power and retries.
3. **Msg3 – scheduled transmission.** The UE applies the TA and sends, on PUSCH with HARQ, its identity and request – for initial access the RRC Connection Request with S-TMSI or a random value.
4. **Msg4 – contention resolution.** The eNB echoes the identity it decoded. A UE whose identity matches has won; the temporary C-RNTI becomes its C-RNTI. Others start the backoff and retry.

Two UEs that chose the same preamble in the same occasion both receive the same RAR and both send Msg3 on the same resources; typically only one Msg3 is decoded (capture effect), which is why Msg4 must echo the identity.

5.4 Contention-free random access

For handover the target cell reserves a preamble and includes it in the handover command; for downlink data arrival the eNB signals it on the PDCCH. The UE sends the dedicated preamble, receives the RAR with TA and grant, and the procedure ends – Lecture 4 relied on this for the access step of X2 handover.

5.5 RACH overload and its control

When many devices access simultaneously – after a power outage, at a stadium, or when thousands of meters report at the same time – collisions multiply, retries multiply further, and the PRACH collapses. LTE offers several countermeasures:

- **Access class barring (ACB)**: SIB2 broadcasts a barring factor a and time; before accessing, a UE draws a uniform number and proceeds only if it is below a, otherwise it waits. Extended access barring (EAB) targets delay-tolerant devices.
- **Backoff indicator** in the RAR: losers wait a random time up to the indicated maximum.
- **More PRACH resources**: additional PRACH occasions per frame, at the cost of uplink capacity.
- **Separate resources for IoT**: NB-IoT and LTE-M define their own random access channels.

5.6 Random access in 5G NR

NR keeps the four-step structure but adapts it to beams and to low latency [5]: each synchronisation signal block (SSB, one per beam) is associated with specific PRACH occasions, so the preamble also tells the gNB which downlink beam the UE sees best; preambles of length 839 or 139 with several subcarrier spacings support everything from long-range rural cells to mmWave; and Release 16 adds **2-step RACH**, in which MsgA combines the preamble and the Msg3 payload and MsgB combines the RAR and contention resolution, halving the round trips.

6 Mathematical models

6.1 Timing advance

Problem. How precisely can LTE align a UE, and how far can it be?

Model. The RAR carries $T_A \in \{0, \ldots, 1282\}$; the UE advances its uplink by $N_{\mathrm{TA}} = 16\, T_A\, T_\mathrm{s}$ with $T_\mathrm{s} = 1/(15000 \times 2048)$ s [3].

Derivation. One step corresponds to a round-trip time of $16 T_\mathrm{s} = 0.521\,\mu$s and therefore to a one-way distance

$$\Delta d = \frac{c \cdot 16\, T_\mathrm{s}}{2} = \frac{2.998 \times 10^8 \times 0.521 \times 10^{-6}}{2} \approx 78.1\,\mathrm{m}, \qquad d_{\max} = 1282\, \Delta d \approx 100\,\mathrm{km}. \tag{2}$$

Interpretation. The step is a fraction of the normal cyclic prefix (4.7 μs), so residual misalignment is absorbed by the CP. Compare with GSM's 553 m step (Lecture 1): the finer step reflects the much shorter LTE symbol and the need to keep all users within the CP. After access, MAC TA commands (6 bits, relative) keep the alignment as the UE moves.

6.2 Cyclic shift and cell radius

Model. The sequence duration $T_{\mathrm{SEQ}} = 800\,\mu$s spans $N_{\mathrm{ZC}} = 839$ samples, so one sample lasts $T_{\mathrm{SEQ}}/N_{\mathrm{ZC}} = 0.954\,\mu$s. A UE at distance r shifts its preamble by the round-trip delay $2r/c$; with multipath delay spread τ_{ds} the received energy occupies $2r/c + \tau_{\mathrm{ds}}$.

Condition. To be unambiguous, this must fit inside one cyclic-shift zone:

$$N_{\mathrm{CS}}\,\frac{T_{\mathrm{SEQ}}}{N_{\mathrm{ZC}}} \;\geq\; \frac{2r}{c} + \tau_{\mathrm{ds}} \quad\Longrightarrow\quad r_{\max} = \frac{c}{2}\left(N_{\mathrm{CS}}\,\frac{T_{\mathrm{SEQ}}}{N_{\mathrm{ZC}}} - \tau_{\mathrm{ds}}\right). \tag{3}$$

Example and interpretation. With $\tau_{\mathrm{ds}} = 5.2\,\mu$s: $N_{\mathrm{CS}} = 13$ gives $r_{\max} \approx 1.1$ km and $\lfloor 839/13 \rfloor = 64$ preambles from a *single* root; $N_{\mathrm{CS}} = 119$ gives $r_{\max} \approx 16.2$ km but only 7 preambles per root, so $\lceil 64/7 \rceil = 10$ roots are needed. Different roots are not orthogonal (cross-correlation $1/\sqrt{839} \approx 0.035$), so large cells have more false alarms and inter-cell preamble interference – a coverage/robustness trade-off planned per cell. The formula ignores a small detection margin used in practice; the guard-time limit of Table 1 must also hold.

6.3 Collision probability

Model. n UEs contend in the same PRACH occasion and choose uniformly among M contention preambles (e.g. $M = 54$ when 10 of the 64 are reserved for CFRA). A UE succeeds if no other UE chose its preamble (ignoring detection errors and Msg3 capture).

Derivation. The probability that a given UE is alone on its preamble is

$$P_{\mathrm{s}}(n) = \left(1 - \frac{1}{M}\right)^{n-1}, \tag{4}$$

and the expected number of successful UEs in the occasion is

$$S(n) = n\left(1 - \frac{1}{M}\right)^{n-1}. \tag{5}$$

Treating n as continuous, $\mathrm{d}S/\mathrm{d}n = 0$ gives $n^{*} = -1/\ln(1 - 1/M) \approx M$, where $S(n^{*}) \approx M/\mathrm{e}$ – exactly the slotted-ALOHA efficiency of $1/\mathrm{e} \approx 37\%$ per preamble.

Interpretation. For $M = 54$: 10 contenders succeed individually with 84.5%; 54 contenders give about 20 successes (37%); 150 contenders give only 9.3 successes – *fewer* than 54 did (Figure 1). Beyond the optimum, more load means less throughput, and the failed UEs retry, pushing the load further: a positive feedback that leads to collapse unless access is throttled (ACB, backoff). The model ignores power ramping, capture and the multiple PRACH occasions per frame.

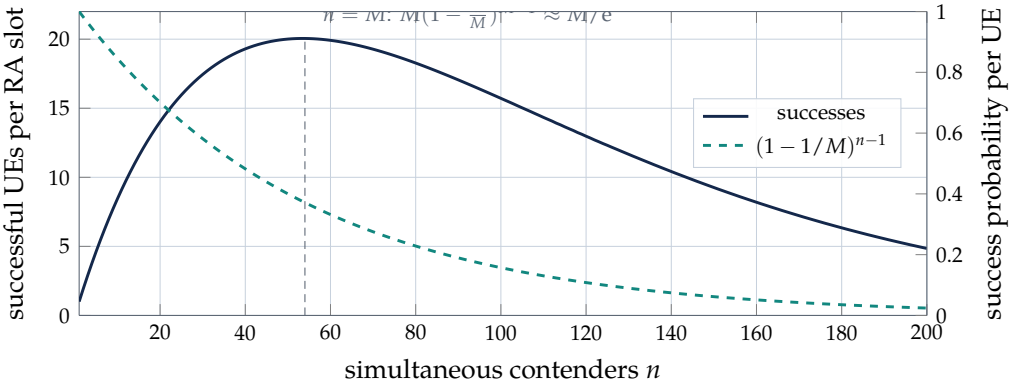


Figure 1. Expected successful UEs per PRACH occasion (left axis, Eq. (5)) and per-UE success probability (right axis, Eq. (4)) for $M = 54$ contention preambles.

6.4 Access delay

If each attempt succeeds independently with probability P_s, the number of attempts is geometric with mean $1/P_s$, and the mean access delay is

$$\bar{D}_{RA} \approx T_{RA} + \left(\frac{1}{P_s} - 1\right)(T_{fail} + \bar{T}_{BO}), \tag{6}$$

where T_{RA} is the duration of a successful 4-step exchange, T_{fail} the time to detect a failure (end of the RAR window or of the contention-resolution timer) and $\bar{T}_{BO}$ the mean backoff (Exercise 4).

7 Network architecture and diagrams

Random access is a purely radio-level procedure between UE and eNB (MAC and PHY), but its outcome feeds the higher layers: Msg3 carries the first RRC message, and Msg4 (in initial access) the RRC Connection Setup, after which the NAS Service Request or Attach Request reaches the MME (Lecture 6). Figure 2 shows the format-0 preamble, Figures 3 to 5 the procedures.

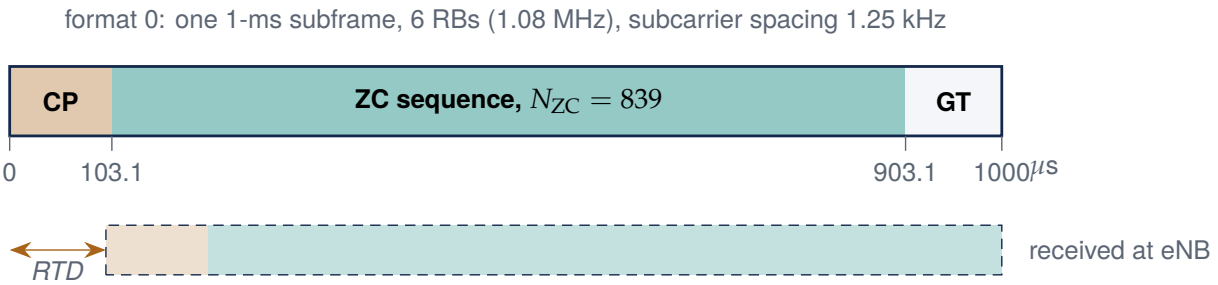


Figure 2. Preamble format 0. The cyclic prefix absorbs the round-trip delay (RTD) and delay spread; the guard time keeps a late preamble from overlapping the next subframe.

8 Step-by-step procedures

Procedure 1 • Contention-based random access for initial access

1. Read SIB2: PRACH configuration index, root sequence, N_{CS}, preamble groups, power parameters, RAR window, contention-resolution timer, ACB parameters.
2. Pass the access-barring check (if configured), select group A or B, pick a preamble at random and transmit it with initial power $P = \min\{P_{max},\ P_{target} + L_p\}$.
3. Monitor the PDCCH for the RA-RNTI during the RAR window. No matching RAR: increase power by the ramping step, apply backoff and go back to step 2 (up to the maximum number of attempts).
4. On RAR: apply TA, store the temporary C-RNTI, send Msg3 on the granted PUSCH, start the contention-resolution timer.
5. On Msg4 with matching identity: success, C-RNTI confirmed, RRC continues. Timer expiry or mismatch: backoff and retry.

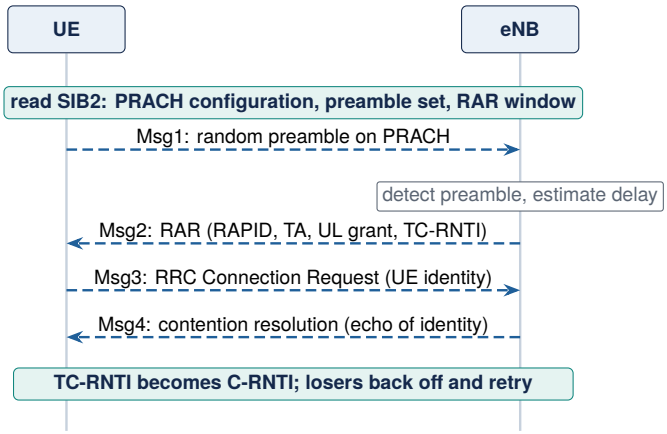


Figure 3. Contention-based (4-step) random access (Procedure 1).

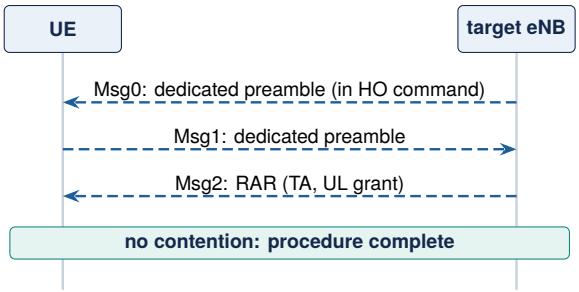


Figure 4. Contention-free random access, e.g. at handover.

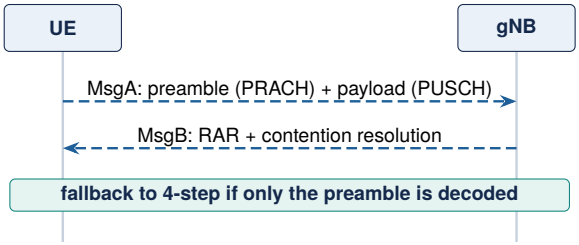


Figure 5. NR 2-step random access (Release 16).

9 Worked examples

Worked example 1 • Timing advance from a measured delay

Problem. The eNB measures a preamble round-trip delay of 21.4 μs. Which T_A is sent and how far is the UE?
Solution. $T_A = \text{round}(21.4/0.521) = 41$; $d \approx 41 \times 78.1 = 3.2\,\text{km}$ (directly: $c\tau/2 = 3.21\,\text{km}$).

Worked example 2 • Planning N_{CS}

Problem. A rural cell must serve UEs up to 12 km with a delay spread of 5.2 μs. Find the smallest N_{CS}, the preambles per root and the number of roots.
Solution. Equation (3): $N_{CS} \geq (2 \times 12\,000/c + 5.2 \times 10^{-6})/0.954 \times 10^{-6} = (80.1 + 5.2)/0.954 = 89.4$. The standard set of N_{CS} values contains 93, which gives $\lfloor 839/93 \rfloor = 9$ preambles per root and $\lceil 64/9 \rceil = 8$ roots.

Worked example 3 • Event-driven IoT burst

Problem. After a power cut, 5000 meters in one cell try to access within 1 s. The cell has one PRACH occasion per 10 ms frame (100 per second) and $M = 54$ contention preambles. Estimate the successes if every meter tries once in a random occasion, and suggest an access-barring factor.
Solution. 50 contenders per occasion on average: $S = 50\,(53/54)^{49} \approx 50 \times 0.40 = 20.0$ successes per occasion, about 2000 in the first second; the 3000 losers retry and add to the next second's load. If the 5000 meters instead all fell into 10 occasions (one per 100 ms), there would be 500 contenders per occasion and $S = 500\,(53/54)^{499} \approx 0.04$: almost nobody succeeds. A barring factor $a \approx 54/500 \approx 0.1$ brings the effective load back to the optimum $n \approx M$, where about $M/\mathrm{e} \approx 20$ UEs succeed per occasion.
Interpretation. In an overloaded random access channel, throttling *increases* the throughput – the central lesson of ALOHA.

10 Technology comparison

Trade-offs. Longer preambles and larger cyclic shifts extend range but reduce the number of orthogonal preambles per root; more PRACH occasions reduce collisions but consume uplink capacity; 2-step RACH halves latency but needs PUSCH resources for payloads that may collide.

11 Key takeaways

Table 2. Random access across generations.

	GSM	LTE	NR
Channel	RACH burst on a timeslot	PRACH, 6 RBs, ZC preambles	PRACH tied to SSB beams
Contention resolution	SDCCH assignment + identity	Msg3/Msg4 with identity	4-step or 2-step (MsgA/MsgB)
TA step / range	553 m / 35 km	78 m / 100 km	depends on numerology (finer at higher SCS)
Contention-free	no	yes (handover, DL data)	yes, plus beam-failure recovery
Overload control	access classes	ACB, EAB, backoff	unified access control

Key takeaways

- Random access provides uplink time alignment and a first identity/grant; it is used for initial access, re-establishment, handover and out-of-sync data.
- Preambles are ZC sequences ($N_{ZC} = 839$): constant amplitude, zero cyclic autocorrelation; 64 per cell from cyclic shifts of one or more roots.
- $r_{max} = \frac{c}{2}(N_{CS}T_{SEQ}/N_{ZC} - \tau_{ds})$; the guard time also bounds the range (format 0: 14.5 km).
- TA step $16T_s = 0.52\,\mu$s $\Rightarrow$ 78 m; 11-bit RAR TA $\Rightarrow \approx 100$ km.
- CBRA: preamble $\rightarrow$ RAR $\rightarrow$ Msg3 $\rightarrow$ contention resolution; CFRA ends after the RAR.
- Successes per occasion $n(1-1/M)^{n-1}$ peak at $\approx M/\mathrm{e}$; overload collapses throughput, so throttle (ACB, backoff).
- NR adds beam-associated PRACH and 2-step RACH.

12 Exercises

Exercise 1 • Level 1 – conceptual • Why not just PUSCH?

Explain what would go wrong if an idle UE transmitted its first message directly on PUSCH without random access. Consider both timing and identity.

Exercise 2 • Level 1 – conceptual • CBRA or CFRA?

For each case state whether CBRA or CFRA is used and why: initial attach; handover; RRC re-establishment after RLF; downlink data arrival for an out-of-sync connected UE.

Exercise 3 • Level 2 – numerical • Format choice

A cell must cover 60 km over flat terrain. Which LTE preamble formats are possible from the guard-time point of view, and what is the smallest N_{CS} (with $\tau_{ds} = 5.2\,\mu s$)? Is it possible with an 800 μs sequence?

Exercise 4 • Level 2 – numerical • Access delay

With $T_{RA} = 20$ ms, $T_{fail} = 15$ ms and a uniform backoff in $[0, 20]$ ms, compute the mean access delay from Eq. (6) for $n = 10$ and $n = 54$ contenders ($M = 54$).

Exercise 5 • Level 3 – analytical • ZC properties

Prove that the periodic autocorrelation of Eq. (1) is zero for every non-zero shift τ when N_{ZC} is prime. (Hint: write $x_u(n+\tau)x_u^*(n)$ and sum a geometric series in n.)

Exercise 6 • Level 3 – analytical • Stability of retries

New UEs arrive at a mean of λ per PRACH occasion; each failed UE retries in the next occasion. Using Eq. (5) with Poisson approximation $S \approx Ge^{-G/M}$ for offered load G per occasion, write the equilibrium condition, find the maximum sustainable λ and discuss what happens beyond it.

Solutions

Solution to Exercise 1

Timing: the UE knows only the downlink timing; its uplink would arrive late by the round-trip delay (up to hundreds of μs), overlapping other users' SC-FDMA symbols and destroying orthogonality. Identity: the eNB has no C-RNTI for the UE and cannot address a grant to it; unscheduled PUSCH resources do not exist. Random access solves both with a robust, delay-tolerant preamble.

Solution to Exercise 2

Initial attach: CBRA (the network does not know the UE). Handover: CFRA (the target reserves a preamble, saving the contention step and time). RRC re-establishment: CBRA (unplanned). Downlink data for an out-of-sync UE: CFRA (the eNB knows the UE and signals a dedicated preamble on the PDCCH), falling back to CBRA if no dedicated preamble is available.

Solution to Exercise 3

The guard time must cover the round trip $2 \times 60\,000/c = 400\,\mu$s: formats 1 (515.6 μs) and 3 (715.6 μs) qualify; 0 and 2 do not. Equation (3) requires $N_{CS} \geq (400 + 5.2)/0.954 = 425$ samples. The largest non-zero value of the standard unrestricted set is 419 ($r_{max} \approx 59$ km), just short; the remaining option is $N_{CS} = 0$ (no cyclic shift), in which each root provides a single preamble and 64 roots are used. This works for any delay, but the 64 preambles are no longer orthogonal (only their low cross-correlation separates them), so detection performance degrades – the price of very large cells. The 800-μs sequence length is not the issue; repetition (format 3) helps the link budget only.

Solution to Exercise 4

$\bar{T}_{\mathrm{BO}} = 10\,\mathrm{ms}$. $n = 10$: $P_{\mathrm{s}} = 0.845$, $1/P_{\mathrm{s}} - 1 = 0.183$, $\bar{D} = 20 + 0.183 \times 25 = 24.6\,\mathrm{ms}$. $n = 54$: $P_{\mathrm{s}} = 0.371$, $1/P_{\mathrm{s}} - 1 = 1.70$, $\bar{D} = 20 + 1.70 \times 25 = 62.4\,\mathrm{ms}$ (optimistic, because retries also add load).

Solution to Exercise 5

$x_u(n+\tau)x_u^*(n) = \exp(-\mathrm{j}\pi u[(n+\tau)(n+\tau+1) - n(n+1)]/N) = \exp(-\mathrm{j}\pi u[\tau^2 + \tau]/N)\exp(-\mathrm{j}2\pi u\tau n/N)$. The first factor is constant in n; summing the second over $n = 0, \ldots, N-1$ gives $\sum_n \mathrm{e}^{-\mathrm{j}2\pi u\tau n/N} = 0$ unless $u\tau \equiv 0 \pmod N$. With N prime and $1 \le u, \tau < N$, $u\tau$ is never a multiple of N, so the autocorrelation is zero for all $\tau \neq 0$. (Periodicity of the quadratic phase modulo N holds for odd N.)

Solution to Exercise 6

In equilibrium the success rate equals the arrival rate: $\lambda = Ge^{-G/M}$. The right-hand side is maximal at $G = M$ with value M/e, so $\lambda_{\max} = M/\mathrm{e} \approx 19.9$ UEs per occasion for $M = 54$. For $\lambda < M/\mathrm{e}$ there are two equilibria: the low-load one ($G < M$) is stable, the high-load one is not. For $\lambda > M/\mathrm{e}$ no equilibrium exists: the backlog grows without bound and successes fall towards zero – the instability that ACB and randomised backoff are designed to prevent.

13 Further reading

The MAC random access procedure is in [1], preamble generation and formats in [2], timing advance in [3], and ACB in the RRC specification [4]. NR random access, including 2-step RACH, is summarised in [5]. Sesia et al. [7] contains a detailed chapter on PRACH design and ZC sequences; Dahlman et al. [6] covers the NR procedures.

References

[1] 3GPP, *Evolved universal terrestrial radio access (E-UTRA); medium access control (MAC) protocol specification*, TS 36.321, 3rd Generation Partnership Project. [Online]. Available: https://www.3gpp.org/DynaReport/36321.htm.

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